

Executive Summary

In this project, I analysed how the massive October crafting update affected the CS2 economy. My main goal was to test a specific business hypothesis: did the new crafting mechanics cannibalise the old case unboxing market or if it actually boosted the entire ecosystem? I built a complete data pipeline and applied statistical analysis to find an answer.

Business Context

Counter-Strike 2 (CS2) has a massive global P2P economy that works like a real-world exchange market. In this project, I treat digital items (skins and cases) as real financial assets. My goal was to use raw market data to analyse how a major product update changed user spending behaviour and the platform's total revenue.

Methodology & Tech Stack

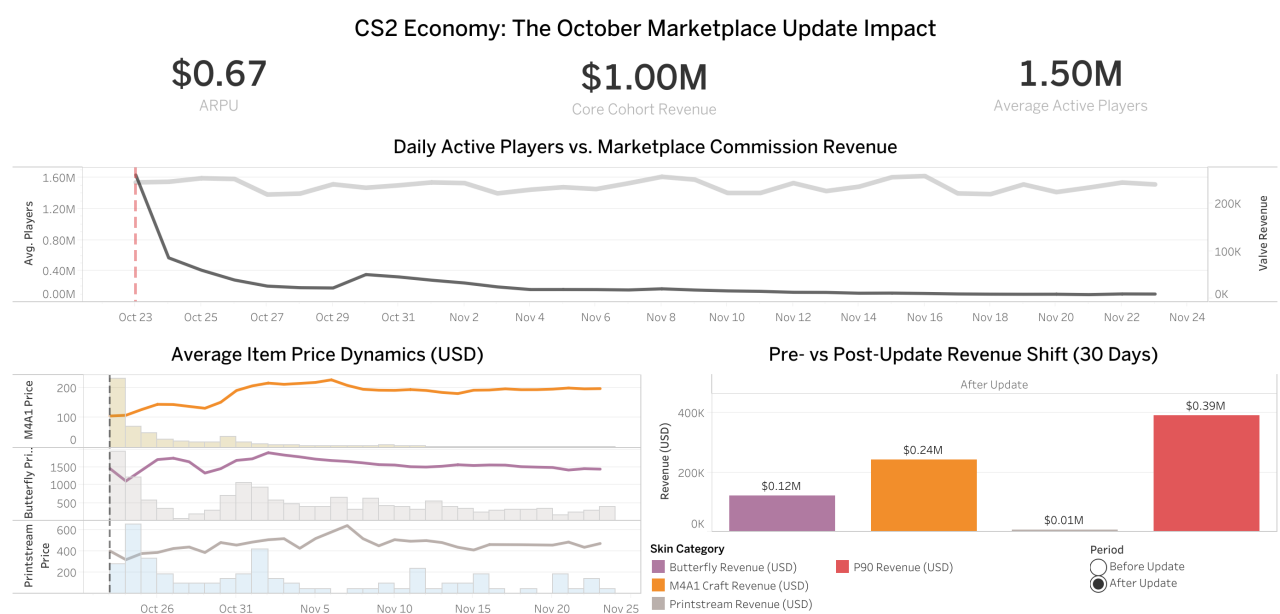
I built this project from scratch, moving from raw API data to the final dashboards. The process started with a custom Python script, utilising pandas and SQLAlchemy, to pull raw JSON data directly from the Steam API. After cleaning the data and handling timezone discrepancies and duplicate transactions, I structured and stored the final datasets in a PostgreSQL database.

To prove that the observed economic shifts were caused by the update and not just random market trends, I applied a Difference-in-Differences approach, tracking the Printstream collection as a baseline control item. And finally, I connected the database to Tableau to build interactive dashboards showing revenue and volume shifts.

Key Findings

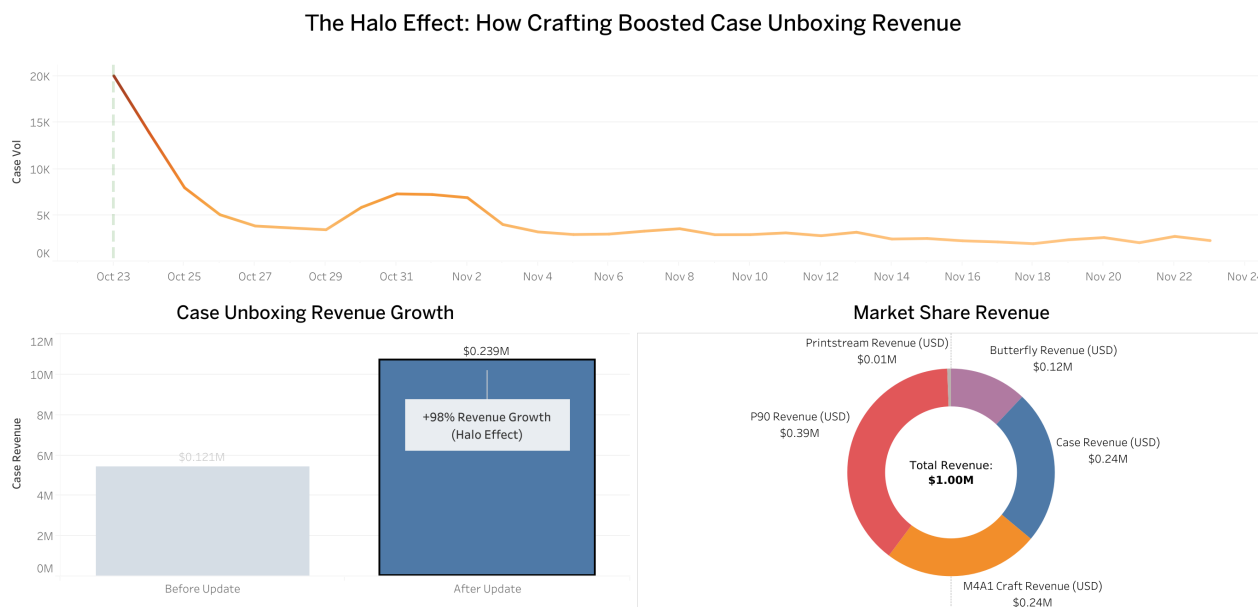
A \$1M Revenue Spike in the Core Cohort

After the update, the items used for crafting, such as the P90 and M4A1, suddenly became incredibly valuable. Focusing strictly on our analysed asset sample, this shift alone generated \$1M in marketplace commissions post-update. Items that were previously stagnant suddenly captured the vast majority of the cohort's transaction volume, becoming the primary drivers of the economy.



The Halo Effect on the Legacy Market

I initially hypothesised that the new crafting system might kill case unboxing. However, the data completely disproved this. Interestingly enough, the player count remained relatively stable, yet ecosystem-wide case unboxing revenue increased by 98%. This shows a powerful "Halo Effect" - the update didn't necessarily bring in a massive amount of users, but it drastically increased the economic engagement and spending behaviour of the existing player base.



Luxury Market Dynamics and Panic Selling

High-tier items, like the Butterfly Knife, experienced a sudden and temporary price drop. Based on the data, this anomaly was caused by a supply shock. Players realised the update introduced a new, predictable generation source for knives, breaking the artificial shortage. Fearing a permanent loss of value of their luxury assets, some users panic-sold their knives at a discount to cash out and secure their funds before the market crashed further.

Conclusion

The October update was a massive success for the CS2 marketplace. My analysis demonstrates that introducing a well-designed resource sink did not cannibalise existing revenue streams. Even without a significant spike in the overall player base, the update successfully monetised stagnant assets and drove massive financial engagement, proving that strategic economic mechanics can elevate an entire virtual market.